Answers and talk notes

Math is Playful

What your student is learning

Tonight we play with the Tower of Hanoi puzzle and look for a pattern that predicts the number of moves.

Answers

Ask your student to explain before you look.

1. The Tower of Hanoi is a puzzle with discs, three spots, and rules.
2. The moves 1, 3, 7 for one, two, and three discs follow a pattern.
3. Double the moves and add 1 is a pattern rule. 3 doubled is 6, plus 1 is 7.
4. A good prediction uses the pattern to say what comes next before you move.
5. A table with discs in one column and moves in the other organizes the results.

Talk about it

What is the smallest tower you could solve first?

Home mission

Use three coins of different sizes and three spots on the table. Move one coin at a time, and never put a big coin on a small one. Start with one coin, then two. Ask your student to predict before moving three.

When they are stuck, ask

- What happened with just two discs?
- How does each answer relate to the one before it?
- Can you predict before you play?

Try not to say

- Just keep moving pieces until it works.
- There is no pattern, it is just a game.
- Let me solve it for you.